

# David Parkhurst

Outcome Engineer. Founder & Architect, Aitherium / AitherOS. USAF veteran.

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I run a software factory, not a backlog. Over 14 months I built AitherOS — an AI-native operating platform where 33 specialised agents plan, code, review, audit, deploy and write about their own work — and used it to land 21,065 commits, 642 self-testing quality gates, ~1,200 MCP tools and 270 published posts, most of it authored and shipped by agents under programmatic governance I designed. Before that: 12 years racking, securing and automating enterprise infrastructure, including the second-largest Active Directory on earth for the US Air Force. I have scars from every layer of the stack and a documented, measured opinion about how autonomous systems fail. Onebrief's Outcome Engineering charter — orchestration, governance, memory, evaluation, self-healing, ship — is the job I have been doing, in the open, every day.

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|-----------------------------------------|----------------------------------------|------------------------------------------|-----------------------------------|---------------------------------|-----------------------------------------|
| <b>21,065</b><br>commits since Jun 2025 | <b>13,928</b><br>commits, last 90 days | <b>642</b><br>self-testing quality gates | <b>~1,200</b><br>MCP tools served | <b>369</b><br>services declared | <b>270</b><br>posts published by agents |
|-----------------------------------------|----------------------------------------|------------------------------------------|-----------------------------------|---------------------------------|-----------------------------------------|

## OUTCOME ENGINEERING, BY THE POSTING'S OWN LIST

### ARCHITECT MULTI-AGENT SYSTEMS

Designed the AitherOS dispatch ladder: a 33-agent roster (Demiurge codes, Hydra reviews, Athena audits, Apollo profiles, Prometheus deploys, Scribe documents) behind a single MCP gateway, with capability-based routing, swarm coding (11 parallel LLM agents), agent-to-agent mesh messaging (awrelay), and a ratchet loop (propose -> apply -> score -> keep/revert) driving measurable improvement. All LLM traffic routes through one priority scheduler (MicroScheduler) across a local RTX 5090, a DGX Spark pool and rented cloud GPUs.

### ENGINEER CONTEXT AND MEMORY

Built the semantic substrate agents work from: a code graph (awgraph, AST + tree-sitter call graphs, recall@25 measured), a repo-understanding layer (repowise), a portable scoped memory (awm), a knowledge graph with hybrid search (Nexus), a 25-question retrieval gold set with canaries that gate every chunking/embedding change, and a per-project doctrine file bind-mounted into every agent workspace. Trained a code-search embedder 13x smaller than its teacher that beats it, for \$3 of GPU.

### DESIGN SELF-HEALING LOOPS

Autoheal + a death watchdog (opt-out) restart wedged services; a 4-hourly probe runs 79 live gates in-fleet and pages on VIOLATION and on DEAD separately; a data-writability gate auto-repairs non-sensitive permission drift and quarantines anything credential-shaped; a paging-path gate asserts the page itself LANDS by round-trip, after an outage where detection worked and the page died. Root-caused and documented the incidents (nine-hour WSL flap,

### IMPLEMENT AUTOMATED GOVERNANCE

Replaced manual review with 642 checker gates, each required to carry a --self-test proving it can still fail, each wired into one of four unattended lanes (CI, 4-hour in-fleet probe, host probe, self-test-only with a stated reason). A ledger rule forbids recording a mechanically detectable defect as a note: it must become a check. Stop hooks block a session that changed code without touching a gate or the ledger. A written "no false blockers" doctrine defines exactly three legitimate reasons an agent may stop (physically impossible, irreversible and destructive, owner said no) and measurably cut stalled loops.

### BUILD RIGOROUS EVALUATION FRAMEWORKS

Every gate exits 0 / 1 / 2 (pass / violation / could-not-judge) and "could not judge" is never a pass. Built the gauntlet: capture -> change -> recapture -> blind A/B verdict -> raw-exit-code gate -> accept/reject with a rejected ledger. Caught a trainer that reported "3 epochs over 20,000 records" while touching 192 rows, an optimiser that had never run a round while logging "completed", and a mock-driven test suite that never executed the loop it tested. Bucketed metrics by seen/unseen so an aggregate at ceiling cannot hide a bucket at floor.

### PROTOTYPE AND DEPLOY

Ship cadence: a new tenant app (backend, agent brain pack, portal, Cloudflare tunnel, GitHub Pages) in about ten minutes; three paying or pilot tenant appliances live (repair shop, game studio, political campaign tooling); 61 public repos; four packages on PyPI; an on-device 1.7B model running in a Pixel 10 browser talking to the platform; a WebMCP design studio in the browser; and every blog post converted to a narrated

thundering-herd boot storm, silent-empty SSE lanes) as gates, not memories.

video by the pipeline that made the one attached to this application.

## EXPERIENCE

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### **Founder, Architect (AitherOS, the aw\* stack, awnix)** — Aitherium

Jul 2025 - Present · Remote

- Architected AitherOS: 369 declared services across 12 layers, run as podman quadlets on Rocky/Debian, with Postgres replication, Redis Sentinel, Cloudflare tunnels, and GPU scheduling across local, DGX and rented (Lambda, Vast, Hetzner, AWS) capacity.
- Built the aw\* (Aither World) family: 64 registered bricks, 40 public, composed into 14 stacks. awnix is an immutable Linux you can hand to an agent; awdk builds agent fleets in three lines; awgit is op-log-aware git for many agents in one tree; awgraph, awm, awrelay, awiam, awbac, awseal, awdit cover code graph, memory, messaging, identity, RBAC, signing and audit. Four on PyPI.
- Governance as code: 642 self-testing gates, a per-gate lane manifest (CHK001-006) that asserts 100% coverage and refuses a gate that runs nowhere, and a debt-ledger protocol where a detectable defect must become a check. Measured the ledger at 79 rows/day in vs 4.2/day out before the rule; it stopped being a write-only archive after.
- 21,065 commits since June 2025 (13,928 in the last 90 days), 33,075 tracked files, 270 blog posts, 160 scheduled routines - the majority authored by agents under those gates, with every claim required to carry its command and exit code.
- Trained and shipped models: a code-search embedder 13x smaller than its teacher that beats it (recall measured on the platform's own corpus); Gemma-4 and orchestrator fine-tunes with a positive-baseline rule (a trainer must beat a named baseline or it is not a result); Bonsai 1.7B running in-browser on a phone.
- Security posture designed fail-closed: authz gates deny on every non-happy path, no caller-supplied tenant ids in a decision, dual-key rotation of the fleet credential across 43 verifier sites, secrets vault with audit round-trip asserted by a gate.
- Sold and operated real tenants: J Games & Repair, Digital Ground Game, Garg - each a generated repo, appliance image, brain pack and portal the customer owns.

### **Senior Enterprise Services Engineer (prev. Escalation Engineer, Senior TAM, ESE)** —

Jan 2022 - Present · Remote

Tanium

- Platform SME on the hardest escalations for enterprise and federal endpoint-management customers; 195 cases in six months at 100% CSAT.
- Rewrote the Tanium Risk Assessment tooling in Python to scale from 500 to unbounded endpoints; applied to a 50,000-endpoint environment.
- Built Client Health dashboards and automation moving customers toward a self-healing endpoint estate; authored the guide and ran the live webinar (hundreds of attendees).
- Fully automated a home lab through Tanium + PowerShell + Python as a self-managing environment - the seed of AitherOS.
- 2x Star Performer award (2024).

### **HPC Linux System Engineer** — Boeing

Apr 2020 - Jan 2022 · Seal Beach, CA

- Owned the enterprise tape archive (Oracle HSM / SAM-QFS -> Varsity Storage Manager as technical lead) and a fibre-channel SAN for HPC.
- Wrote Python automation for recovery, monitoring and archival; led LTO6 -> LTO8 migration and data-at-rest encryption (OKM 3, IBM SKLM).

### **Enterprise Active Directory Engineer, Functional Lead, Supervisor Directory Services Ops** — United

Jan 2014 - Jan 2020 · Honolulu HI; Lajes Field, Portugal; Keesler AFB

States Air Force - Cyber Systems Operations

- Led Tier-2 operations for a \$10B, 850,000-user, 690-domain-controller, 230-site Active Directory - the second-largest AD network in the world - including classified and unclassified GAL/Exchange services 24/7.
- Automated DC maintenance and health checks for 70+ domain controllers in PowerShell; performed an authoritative ADDS restore recovering 3,000 accounts with zero collateral impact.
- Identified a training gap, wrote the ADDS troubleshooting curriculum, and trained 357 technicians across four organisations.
- Red Flag 17-1 blue-team operator, Superior Performer; Junior Technician of the Year 2017; Information Dominance award 2018; promoted SSgt in the 94th percentile.

## SKILLS

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**Agentic systems:** multi-agent orchestration, MCP (1, 200+ tools served), tool-calling harnesses, agent memory and knowledge graphs, evaluation harnesses, ratchet/keep-revert loops, prompt and context engineering, on-device inference (WebGPU/wasm)

**Models and training:** vLLM, llama.cpp, LoRA/fine-tuning, embedding distillation, KV-cache quantisation, GPU scheduling, DeepSeek/Qwen/Gemma/Bonsai

**Languages:** Python, TypeScript/Node, PowerShell, Bash, SQL

**Platform:** PostgreSQL (replication), Redis Sentinel, Podman/Docker, systemd quadlets, k3s, Next.js/React, FastAPI, Cloudflare (Workers, Tunnels, Pages), GitHub Actions self-hosted fleets, OpenTofu

**Cloud and hardware:** AWS, Lambda, Vast, Hetzner, NVIDIA DGX Spark, RTX 5090, WSL2

**Ops discipline:** fail-closed authz, secret rotation, observability-to-action loops, incident root-cause, tape/SAN/HPC storage, Windows and Linux at 850K-user scale

## CERTIFICATIONS

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Tanium Certified Professional - Endpoint Management (2024-2026) · Tanium Certified Operator (2022-2026) · GIAC Certified Windows Security Administrator (GCWN) · GIAC Certified Enterprise Defender (GCED) · CompTIA Security+

## PROOF

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AitherOS / aw\* org: [github.com/Aitherium](https://github.com/Aitherium) · Blog (270 posts, agent-written and gated): [blog.aitherium.com](https://blog.aitherium.com) · 13x smaller embedder beats its teacher: [blog.aitherium.com/code-search-embedder-13x-smaller-beats-its-teacher](https://blog.aitherium.com/code-search-embedder-13x-smaller-beats-its-teacher) · Live demo hub: [demo.aitherium.com](https://demo.aitherium.com) · Who I am (this site): [wizzense.github.io](https://wizzense.github.io)

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Every figure above was measured on 2026-09-03; the commands are in `facts/measured.md` in the repository behind [wizzense.github.io](https://wizzense.github.io). This document was assembled by the AitherOS agent platform and reviewed by its author.